

§3.1

Measures of Center

One number for the whole data set

Today's Goal

- Calculate the mean, median, mode, and midrange of a dataset
- Determine which measure of center is most appropriate for a given situation
- Understand the concept of resistance and how outliers affect measures of center
- Use technology (Excel) to calculate measures of center

What You Will Be Graded On

LT.6 Numerical Summaries of Data

I understand the different numerical summaries for data (e.g., mean, median, mode, standard deviation, etc.). I can find and interpret the values of these numerical summaries for a given data set.

Nine Commutes, One Number

Nine students' commutes to campus this morning, in minutes.

8 10 12 12 15 18 20 25 95

**What one number is
the typical commute?**

More than one honest answer. Today you get four, and a reason to pick one.

Four Ways to Say “Center”

Definition 3.1.1

A **measure of center** is a value at the “center” of a data set.

We will look at four measures of center:

mean

median

mode

midrange

Add Them All, Divide by How Many

Definition 3.1.2

The **mean** of a set of data is the measure of center found by adding all of the data values and dividing the total by the number of data values.

Sample of size n : $\bar{x}$, read “x-bar”

$$\bar{x} = \frac{\sum x}{n}$$

Population of size N : μ , read “mu”

$$\mu = \frac{\sum x}{N}$$

People call this the *average*. We will not. Why might that word cause trouble?

Computing a Mean by Hand

Seven students from a class of 30 counted the text messages they sent yesterday.

14 22 9 31 22 18 24

Find the mean number of messages.

Add: $14 + 22 + 9 + 31 + 22 + 18 + 24 = 140$

Divide by the count: $\bar{x} = \frac{140}{7} = 20$ messages

Seven of the 30, so this is a sample: $n = 7$ and the symbol is $\bar{x}$.

One Wild Value Changes Everything

Definition 3.1.3

A statistic is **resistant** if the presence of extreme values (outliers) does not cause it to change very much.

Same seven students. One of them mistyped it: the 31 was really 310.

14 22 9 310 22 18 24

New total 419, so $\bar{x} = 419/7 \approx 59.9$ messages. It was 20.

So the mean is not resistant. One value moved it by almost 40.

The Value in the Middle

Definition 3.1.5

The **median** of a data set is the measure of center that is the *middle value* when the original data values are arranged in order of increasing magnitude. So the median will separate the bottom half of the data values from the top half.

Symbols. Sample median: $\tilde{x}$, read “x-tilde,” or just M . Population median: $\tilde{\mu}$, read “mu-tilde.”

“In order” is doing real work in that sentence. Sort first, every time.

Odd Count, Even Count

Procedure 3.1.6 Finding the Median

To find the median of a data set, begin by sorting the data values in increasing order.

- If the number of data values is *odd*, then the median is the data value located in the exact middle of the sorted list.
- If the number of data values is *even*, then the median is found by computing the mean of the two middle values in the sorted list.

Even count: no single middle value, so split the difference.

Finding a Median, Both Ways

The seven text-message counts from Example 1, now sorted:

9 14 18 22 22 24 31

(a) Seven values. What is the median?

Odd count, so the 4th value: $\tilde{x} = 22$ messages.

(b) An eighth student sent 20. Now what is the median?

Sorted: 9, 14, 18, 20, 22, 22, 24, 31. Even count, so

$\tilde{x} = (20 + 22)/2 = 21$ messages.

(c) Back to seven values, with the typo: 31 becomes 310. Median?

Sorted: 9, 14, 18, 22, 22, 24, 310. Still 22. Resistant.

The Commutes, Two Ways

8 10 12 12 15 18 20 25 95

Mean

$$\bar{x} = \frac{215}{9} \approx 23.9 \text{ minutes}$$

Median

$$\tilde{x} = 15 \text{ minutes}$$

**Which one would you tell
your friend? What is the
95 doing to the other?**

The Value That Shows Up Most

Definition 3.1.8

The **mode** of a data set is the value (or values) that occur with the greatest frequency, provided that the frequency is more than 1. A data set with two modes is called **bimodal**; one with more than two modes is called **multimodal**; one with no repeating value has *no mode*.

A tie for most frequent is not “no mode.” Every value in the tie is a mode.

No adding, no sorting. That is why the mode also works for words, not just numbers.

Finding the Mode

(a) Shoe sizes of ten students: 7, 8, 8, 9, 9, 10, 8, 11, 9, 6. What is the mode?

Size 8 appears three times, and so does size 9. Two modes: 8 and 9. The data is bimodal.

(b) Favorite color of twelve students: blue 5, green 3, red 3, purple 1. What is the mode?

Blue. No mean or median is possible here, but the mode is.

(c) Four quiz scores: 3, 5, 8, 12. What is the mode?

Nothing repeats, so there is no mode.

Halfway Between the Extremes

Definition 3.1.10

The **midrange** of a data set is the measure of center that is the value midway between the maximum and minimum values in the original data set.

$$\text{Midrange} = \frac{(\text{max data value}) + (\text{min data value})}{2}$$

Only two values go in. Every other value in the set is ignored.

Midrange Is Not the Median

The nine commute times again, in minutes.

8 10 12 12 15 18 20 25 95

(a) What is the midrange?

$$(95 + 8)/2 = 51.5 \text{ minutes.}$$

(b) The median was 15. Both are a “middle.” Why are they so far apart?

The median is the middle *position* in the sorted list. The midrange is the middle of the *number line*, so the 95 drags it up.

(c) Is the midrange resistant?

No. It is built from the two most extreme values.

Now You Try: All Four

You Try. Hours of sleep for eight students: 6, 7.5, 9, 6, 5.5, 7, 8, 6. Find all four measures of center.

(a) Mean

$$\bar{x} = 55/8 = 6.875 \text{ hours}$$

(b) Median

Sorted: 5.5, 6, 6, 6, 7, 7.5, 8, 9, so $\tilde{x} = (6 + 7)/2 = 6.5$ hours

(c) Mode

6 hours, reported three times

(d) Midrange

$$(9 + 5.5)/2 = 7.25 \text{ hours}$$

Let Excel Do the Arithmetic

With the data in cells B2 through B51, one formula each:

Measure	Formula	In the notes
Mean	<code>=AVERAGE (B2 : B51)</code>	Example 3.1.4
Median	<code>=MEDIAN (B2 : B51)</code>	Example 3.1.7
Mode	<code>=MODE . MULT (B2 : B51)</code>	Example 3.1.9
Midrange	<code>=(MAX (B2 : B51) + MIN (B2 : B51)) / 2</code>	Example 3.1.11

`MODE . MULT` returns every mode, so a bimodal column shows two numbers. The four You Try questions in the notes use these on real data.

When You Only Have the Table

No original values, only classes and frequencies. The best we can do is *estimate* the mean with the class midpoints x and frequencies f :

$$\bar{x} \approx \frac{\sum(f \cdot x)}{\sum f}$$

Each midpoint stands in for every value in its class. That is why the answer is an estimate, not the mean.

Estimating from the Table

Procedure 3.1.12 Estimating the Mean from a Grouped Frequency Distribution

For each class in the grouped frequency distribution table:

1. Find the class midpoint x .
2. Multiply the class midpoint x by the class frequency f to get $f \cdot x$.

Then compute totals across all classes:

1. Sum the frequencies to get $\sum f$.
2. Sum the products to get $\sum (f \cdot x)$.

Estimating from the Table, Continued

Procedure 3.1.12, continued

Finally:

1. Estimate the mean using the formula $\bar{x} \approx \frac{\sum(f \cdot x)}{\sum f}$.

Two products, two sums, one division. The table does the bookkeeping.

Estimating a Mean from Classes

Forty patients recorded how long they waited at a clinic. Estimate the mean wait.

Wait (minutes)	Frequency f	Midpoint x	$f \cdot x$
0–9	6	4.5	27.0
10–19	14	14.5	203.0
20–29	11	24.5	269.5
30–39	6	34.5	207.0
40–49	3	44.5	133.5
Totals	$\sum f = 40$		$\sum(f \cdot x) = 840.0$

$\bar{x} \approx 840.0/40 = 21.0$ minutes. This is an estimate, so write $\approx$ and not $=$. The true mean needs the original values.

When Some Values Count More

Definition 3.1.14

When different data values are assigned different weights, the **weighted mean** is computed using the formula

$$\bar{x} = \frac{\sum (w \cdot x)}{\sum w}$$

where x represents the data values and w represents the corresponding weights.

Divide by $\sum w$, not by how many values there are.

The Steps

Procedure 3.1.15 Computing a Weighted Mean

For each data value:

1. Identify the data value x and its corresponding weight w .
2. Multiply $x \cdot w$ to get the weighted contribution.

Then compute totals:

1. Sum the weights to get $\sum w$.
2. Sum the weighted contributions to get $\sum (w \cdot x)$.

Finally:

1. Compute the weighted mean as $\bar{x} = \frac{\sum (w \cdot x)}{\sum w}$.

GPA Is a Weighted Mean

A student's first-semester grades and units are below. Quality points: A = 4, B = 3, C = 2, D = 1, F = 0. Compute the GPA.

Grade	Points x	Units w	$w \cdot x$
B	3	4	12
A	4	3	12
C	2	3	6
D	1	2	2
Totals		$\sum w = 12$	$\sum(w \cdot x) = 32$

$\bar{x} = 32/12 \approx 2.67$. Divide by $\sum w = 12$, not by 4 courses: the units are the weights.

Which Measure, When?

Measure	Reach for it when	Watch out
Mean	Symmetric data, no outliers.	Not resistant.
Median	Skewed data, or outliers.	Sort first.
Mode	Qualitative data.	May not exist, or may be several.
Midrange	A quick rough number.	Uses only two values.

Rule of thumb. Mean close to the median: roughly symmetric. Mean well above the median: skewed right, since a few large values pull the mean up. Well below: skewed left.

“Typical Pay Here Is \$89,000”

Nine employees. Salaries in thousands of dollars.

38 41 44 46 48 52 55 60 420

The owner says typical pay is \$89,000. Fair?

Mean $804/9 \approx 89.3$. Median 48. Eight of nine earn less than “typical.”

One value skews it. Report the median, and always ask which number you were given.